

4.2. The Vortex Panel Method (Rough Draft)

An airfoil section of arbitrary shape can be synthesized by combining a uniform flow with a curved vortex sheet wrapped around the surface of the airfoil, as shown in Fig. 4.2.1. The strength of this sheet must vary along the surface of the airfoil in such a way that the normal component of velocity induced by the entire sheet and the uniform flow is zero everywhere along the surface of the airfoil. In most cases, the strength distribution necessary to satisfy this condition is very difficult or impossible to determine analytically. However, for numerical computations, such a sheet can be approximated as a series of flat vortex panels wrapped around the surface of the airfoil. In the limit as the length of each panel becomes very small the panel solution approaches that for the curved vortex sheet.

Consider first a single flat vortex panel, of length l_i , extending from the point (x_i, y_i) to the point (x_{i+1}, y_{i+1}) as shown in Fig. 4.2.2. We shall consider here a linear strength distribution along the panel with strength equal to γ_i at point i and strength equal to γ_{i+1} at point $i+1$. In Fig. 4.2.2, two separate coordinate systems are displayed. The x - y coordinate system, called airfoil-coordinates, has its origin at the leading edge of the airfoil with the x -axis pointing along the chord line. The ξ - η coordinate system, denoted as panel-coordinates, has its origin at point i with the ξ -axis pointing along the panel toward point $i+1$. In terms of the ξ - η coordinate system we can express the velocity induced by this vortex panel using Eq. (2.13.9). This gives

$$\begin{Bmatrix} V_\xi \\ V_\eta \end{Bmatrix} = \frac{1}{2\pi l_i} \begin{bmatrix} [(l_i - \xi)\Phi + \eta\Psi] & (\xi\Phi - \eta\Psi) \\ [\eta\Phi - (l_i - \xi)\Psi - l_i] & (-\eta\Phi - \xi\Psi + l_i) \end{bmatrix} \begin{Bmatrix} \gamma_i \\ \gamma_{i+1} \end{Bmatrix} \quad (4.2.1)$$

where

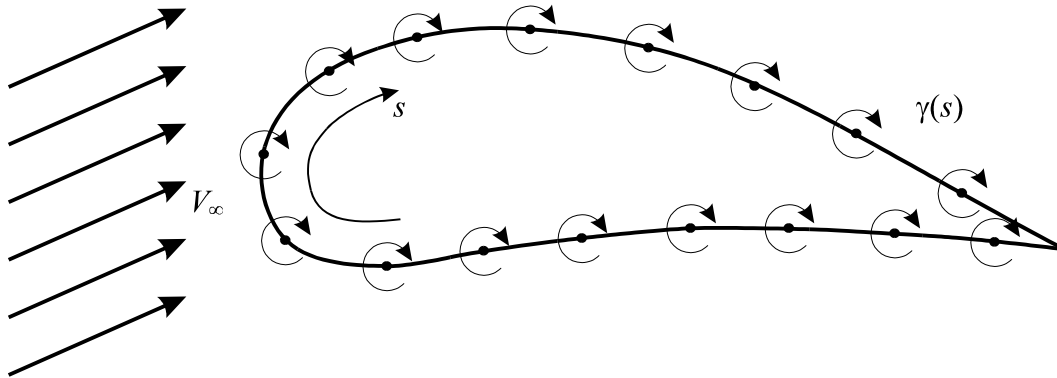


Figure 4.2.1. Synthesis of an arbitrary airfoil section from a uniform flow and a curved vortex sheet distributed over the airfoil surface.

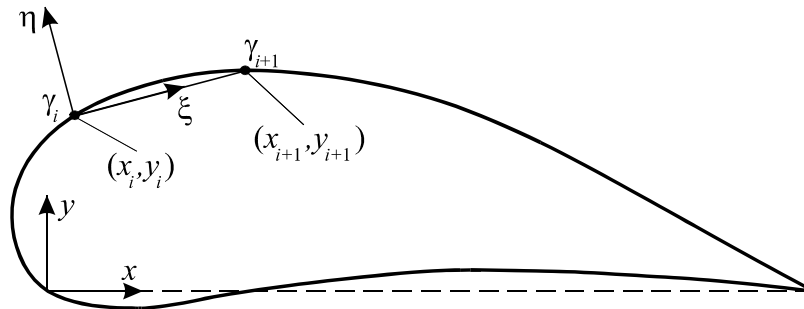


Figure 4.2.2. A vortex panel with linear strength distribution on the surface of an arbitrary airfoil section.

$$\Phi = \tan^{-1} \left(\frac{\eta l_i}{\eta^2 + \xi^2 - \xi l_i} \right) \quad (4.2.2)$$

and

$$\Psi = \frac{1}{2} \ln \left[\frac{\xi^2 + \eta^2}{(\xi - l_i)^2 + \eta^2} \right] \quad (4.2.3)$$

The length of the panel is

$$l_i = \sqrt{(x_{i+1} - x_i)^2 + (y_{i+1} - y_i)^2} \quad (4.2.4)$$

The angle, ϕ , that the panel makes with the x -axis is defined by

$$\cos \phi = \frac{x_{i+1} - x_i}{l_i} \quad (4.2.5)$$

and

$$\sin \phi = \frac{y_{i+1} - y_i}{l_i} \quad (4.2.6)$$

The tangential and normal panel-coordinates, ξ and η , of an arbitrary point can be expressed in term of the axial and normal airfoil-coordinates, x and y , using the transformation,

$$\xi = (x - x_i) \cos \phi + (y - y_i) \sin \phi \quad (4.2.7)$$

$$\eta = -(x - x_i) \sin \phi + (y - y_i) \cos \phi \quad (4.2.8)$$

Using Eqs. (4.2.5) and (4.2.6) in Eqs. (4.2.7) and (4.2.8) we obtain the transformation equation

$$\begin{Bmatrix} \xi \\ \eta \end{Bmatrix} = \frac{1}{l_i} \begin{bmatrix} (x_{i+1} - x_i) & (y_{i+1} - y_i) \\ -(y_{i+1} - y_i) & (x_{i+1} - x_i) \end{bmatrix} \begin{Bmatrix} (x - x_i) \\ (y - y_i) \end{Bmatrix} \quad (4.2.9)$$

The ξ and η components of velocity induced by this linear vortex panel can then be calculated using Eq. (4.2.1). The x and y components of the induced velocity can be calculated from

$$V_x = V_\xi \cos \phi - V_\eta \sin \phi \quad (4.2.10)$$

$$V_y = V_\xi \sin \phi + V_\eta \cos \phi \quad (4.2.11)$$

or after using Eqs. (4.2.5) and (4.2.6),

$$\begin{Bmatrix} V_x \\ V_y \end{Bmatrix} = \frac{1}{l_i} \begin{bmatrix} (x_{i+1} - x_i) & -(y_{i+1} - y_i) \\ (y_{i+1} - y_i) & (x_{i+1} - x_i) \end{bmatrix} \begin{Bmatrix} V_\xi \\ V_\eta \end{Bmatrix} \quad (4.2.12)$$

Combining Eqs. (4.2.12) and (4.2.1), the velocity induced by this vortex panel can be written

$$\begin{Bmatrix} V_x \\ V_y \end{Bmatrix} = [\mathbf{P}] \begin{Bmatrix} \gamma_i \\ \gamma_{i+1} \end{Bmatrix} \quad (4.2.13)$$

where the 2×2 matrix, $[\mathbf{P}]$, will be called the panel coefficient matrix in airfoil coordinates,

$$[\mathbf{P}] \equiv \frac{1}{2\pi l_i^2} \begin{bmatrix} (x_{i+1} - x_i) & -(y_{i+1} - y_i) \\ (y_{i+1} - y_i) & (x_{i+1} - x_i) \end{bmatrix} \begin{bmatrix} [(l_i - \xi)\Phi + \eta\Psi] & (\xi\Phi - \eta\Psi) \\ [\eta\Phi - (l_i - \xi)\Psi - l_i] & (-\eta\Phi - \xi\Psi + l_i) \end{bmatrix} \quad (4.2.14)$$

We can now synthesize an airfoil using $n-1$ panels placed between n nodes on the airfoil surface. The first node is placed at the trailing edge. The nodes are spaced forward along the lower surface, wrapped up around the leading edge, and then spaced back along the upper surface to the trailing edge. The n^{th} node is placed at the trailing edge, at the same location as the first node. The strength of each vortex panel is assumed to be linear along the panel and is required to be continuous from one panel to the next, i.e., the strength at the end of one panel must equal the strength at the beginning of the next panel. The strength is not required to be continuous across the trailing edge, because the sharp corner allows the flow to be singular at this point, i.e., $\gamma_1 \neq \gamma_n$. Using flat panels allows a discontinuity in slope between each panel. This produces weak singularities at the panel intersections. However, this problem is not restrictive if enough panels are used so that the discontinuity in slope is small.

For best results the nodes should be clustered tighter near the leading and trailing edges. The most popular method for attaining this clustering is called “cosine clustering.” For this method we use the change of variables,

$$\frac{x}{c} = \frac{1}{2}(1 - \cos\theta) \quad (4.2.15)$$

Distributing the nodes uniformly in θ will provide the desired clustering in x . If the total **number of nodes is odd**, the nodal coordinates are computed from the algorithm

$$\delta\theta = \frac{\pi}{\text{real}(n/2)} \quad (4.2.16)$$

$$\begin{Bmatrix} x_N(n/2+1) \\ y_N(n/2+1) \end{Bmatrix} = \begin{Bmatrix} 0. \\ 0. \end{Bmatrix} \quad (4.2.17)$$

$$\begin{Bmatrix} x_N(n/2+1+i) \\ y_N(n/2+1+i) \\ x_N(n/2+1-i) \\ y_N(n/2+1-i) \end{Bmatrix} = \begin{Bmatrix} x_u \\ y_u \\ x_\ell \\ y_\ell \end{Bmatrix}, \quad \frac{x}{c} = 0.5\{1 - \cos[\text{real}(i)\delta\theta]\}, \quad i = 1, n/2 \quad (4.2.18)$$

If the **total number of nodes is even**, the nodal coordinates are computed from the algorithm

$$\delta\theta = \frac{\pi}{\text{real}(n/2) - 0.5} \quad (4.2.19)$$

$$\begin{Bmatrix} x_N(n/2+i) \\ y_N(n/2+i) \\ x_N(n/2+1-i) \\ y_N(n/2+1-i) \end{Bmatrix} = \begin{Bmatrix} x_u \\ y_u \\ x_\ell \\ y_\ell \end{Bmatrix}, \quad \frac{x}{c} = 0.5\{1 - \cos[\text{real}(i)\delta\theta - 0.5\delta\theta]\}, \quad i = 1, n/2 \quad (4.2.20)$$

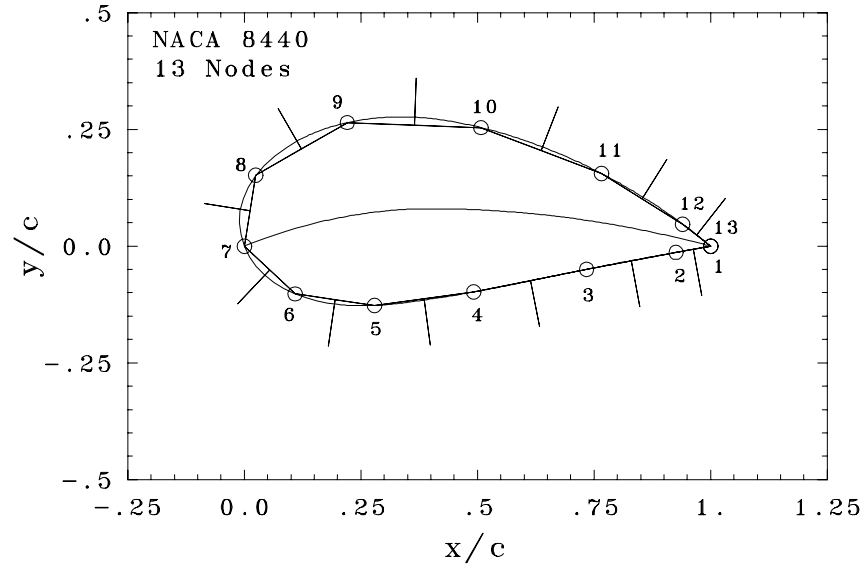


Figure 4.2.3. Vortex panel distribution with cosine clustering and an odd number of nodes.

A nodal distribution using cosine clustering with thirteen nodes is shown in Fig. 4.2.3 and a similar cosine distribution using twelve nodes is shown in Fig. 4.2.4.

To solve for the n unknown strengths, control points are placed at the center of each of the $n-1$ panels. The velocity normal to the panel at each control point, induced by all $n-1$ panels and the uniform flow, is required to be zero. This gives $n-1$ equations for the n unknown nodal strengths.

For the remaining equation, we know that the flow must leave the airfoil from the trailing edge. This means that the velocity just above the trailing edge must equal the velocity just below the trailing edge. If the angle between the upper and lower surface at the trailing edge is non-zero, the trailing edge is a stagnation point. If the angle between the upper and lower surface at the trailing edge is zero, the velocity at the trailing edge can be finite but the velocity must be continuous across the trailing edge. Since there is a discontinuity in tangential velocity

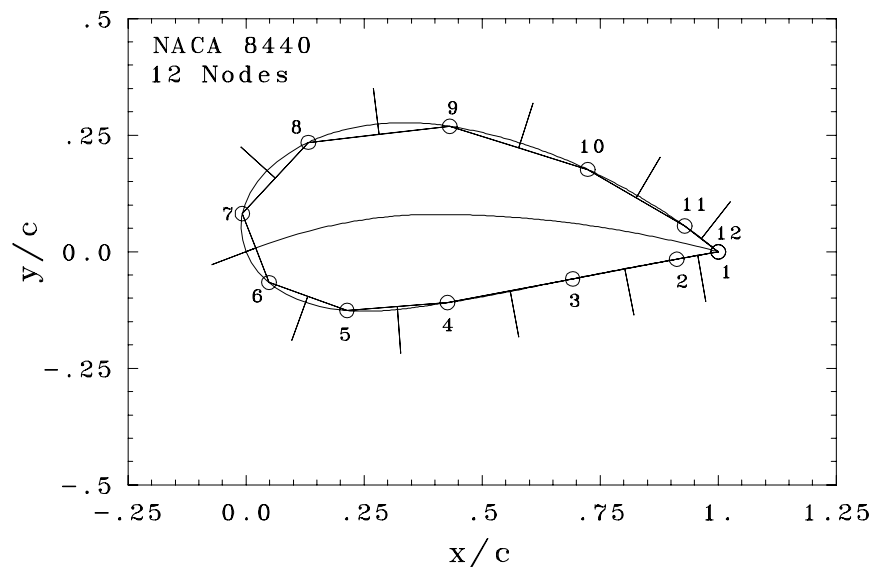


Figure 4.2.4. Vortex panel distribution with cosine clustering and an even number of nodes.

across a vortex sheet, we know that, at the trailing edge, the discontinuity across the upper surface must be equal and opposite to the discontinuity across the lower surface. Thus, the net discontinuity across both surfaces will be zero. Because the discontinuity in tangential velocity across a vortex sheet is equal to the local strength of the sheet, at the trailing edge, the strength of the upper surface must be equal and opposite to the strength of the lower surface. That is

$$\gamma_1 + \gamma_n = 0. \quad (4.2.21)$$

This condition is called the Kutta condition and it provides the remaining equation necessary to solve for the n unknown nodal strengths.

The coordinates of control point i , at the mid point of panel i , are given by

$$\begin{Bmatrix} x_C(i) \\ y_C(i) \end{Bmatrix} = \begin{Bmatrix} \frac{x_N(i) + x_N(i+1)}{2} \\ \frac{y_N(i) + y_N(i+1)}{2} \end{Bmatrix}, \quad i = 1, n-1 \quad (4.2.22)$$

The unit normal to panel i is

$$\mathbf{u}_i = -\sin \phi \mathbf{i}_x + \cos \phi \mathbf{i}_y \quad (4.2.23)$$

or in view of Eqs. (4.2.5) and (4.2.6)

$$\mathbf{u}_i = -\frac{y_{i+1} - y_i}{l_i} \mathbf{i}_x + \frac{x_{i+1} - x_i}{l_i} \mathbf{i}_y \quad (4.2.24)$$

Let the velocity induced at control point i by panel j be, $\mathbf{V}_{ji}$. Then, from Eq. (4.2.13), we can write,

$$\bar{\mathbf{V}}_{ji} = \begin{Bmatrix} V_{xji} \\ V_{yji} \end{Bmatrix} = [\mathbf{P}]_{ji} \begin{Bmatrix} \gamma_j \\ \gamma_{j+1} \end{Bmatrix} = \begin{Bmatrix} P_{11ji} \gamma_j + P_{12ji} \gamma_{j+1} \\ P_{21ji} \gamma_j + P_{22ji} \gamma_{j+1} \end{Bmatrix} \quad (4.2.25)$$

where $[\mathbf{P}]_{ji}$ is the panel coefficient matrix in airfoil coordinates for the velocity induced at control point i by panel j , extending from node j to node $j+1$. Using Eqs. (4.2.24) and (4.2.25), the component of velocity normal to panel i at control point i that is induced by panel j is then,

$$\mathbf{u}_i \cdot \mathbf{V}_{ji} = \left(\frac{x_{i+1} - x_i}{l_i} P_{21ji} - \frac{y_{i+1} - y_i}{l_i} P_{11ji} \right) \gamma_j + \left(\frac{x_{i+1} - x_i}{l_i} P_{22ji} - \frac{y_{i+1} - y_i}{l_i} P_{12ji} \right) \gamma_{j+1} \quad (4.2.26)$$

The component of the free stream velocity that is normal to panel i is

$$\mathbf{u}_i \cdot \mathbf{V}_\infty = V_\infty \left(\frac{x_{i+1} - x_i}{l_i} \sin \alpha - \frac{y_{i+1} - y_i}{l_i} \cos \alpha \right) \quad (4.2.27)$$

Because the total normal velocity at each control point induced by all of the $n-1$ panels plus the uniform flow is required to be zero, we can write

$$\sum_{j=1}^{n-1} \mathbf{u}_i \cdot \mathbf{V}_{ji} = -\mathbf{u}_i \cdot \mathbf{V}_\infty \quad (4.2.28)$$

The $n-1$ equations specified by Eq. (4.2.28), combined with Eq. (4.2.21), is a determinant set which can be solved for the n unknown strengths γ_1 through γ_n . Using Eqs. (4.2.26) and (4.2.27), this can be written in vector form as

$$[\mathbf{A}] \begin{Bmatrix} \gamma_1 \\ \gamma_2 \\ \vdots \\ \gamma_{n-1} \\ \gamma_n \end{Bmatrix} = V_\infty \begin{Bmatrix} [(y_2 - y_1) \cos \alpha - (x_2 - x_1) \sin \alpha] / l_1 \\ [(y_3 - y_2) \cos \alpha - (x_3 - x_2) \sin \alpha] / l_2 \\ \vdots \\ \vdots \\ [(y_n - y_{n-1}) \cos \alpha - (x_n - x_{n-1}) \sin \alpha] / l_{n-1} \\ 0. \end{Bmatrix} \quad (4.2.29)$$

where $[\mathbf{A}]$ is the n by n airfoil coefficient matrix, which can be generated with the following algorithm.

First, set the entire matrix to zero,

$$A_{ij} = 0., \quad i = 1, n; \quad j = 1, n \quad (4.2.30)$$

Next, add the contribution from the first node on each panel,

$$A_{ij} = A_{ij} + \frac{x_{i+1} - x_i}{l_i} P_{21ji} - \frac{y_{i+1} - y_i}{\ell_i} P_{11ji}, \quad i = 1, n-1; \quad j = 1, n-1 \quad (4.2.31)$$

and the contribution from the second node on each panel,

$$A_{ij+1} = A_{ij+1} + \frac{x_{i+1} - x_i}{l_i} P_{22ji} - \frac{y_{i+1} - y_i}{l_i} P_{12ji}, \quad i = 1, n-1; \quad j = 1, n-1 \quad (4.2.32)$$

where the panel coefficient matrix, $[\mathbf{P}]_{ji}$, used in Eqs (4.2.31) and (4.2.32) is for the velocity induced at control point i by panel j , extending from node j to node $j+1$. Finally, add the two terms from the Kutta condition,

$$A_{n1} = 1. \quad (4.2.33)$$

$$A_{nn} = 1. \quad (4.2.34)$$

Notice that, while the strength vector, $\{\gamma\}$, and the right hand vector in Eq. (4.2.29) are both functions of airfoil geometry and angle of attack, the $[\mathbf{A}]$ matrix is a function only of airfoil geometry. Thus, $[\mathbf{A}]$ need only be decomposed once for any given geometry and the strength distribution can be computed from this single decomposed matrix for any angle of attack.

The system of algebraic equations given by Eq. (4.2.29) is a linear system that may be written in the form

$$[\mathbf{A}]\{\gamma\} = \{\mathbf{C}\} \quad (4.2.35)$$

This system is efficiently solved using **LU** decomposition. The details associated with obtaining numerical solutions to linear systems of algebraic equations using **LU** decomposition can be found in any undergraduate text on engineering numerical methods, see for example Hoffman (1992) or Press et al. (1986). It should be noted that the $[\mathbf{A}]$ matrix might contain zeros on the diagonal. Thus, any algorithm used to solve this system must include scaling and pivoting.

Once the strength distribution is known, the velocity at any point in space can be computed by adding the velocity induced by all $n-1$ panels to the free-stream velocity. Thus, the local velocity at the point (x, y) is given by

$$\begin{Bmatrix} V_x \\ V_y \end{Bmatrix} = V_\infty \begin{Bmatrix} \cos \alpha \\ \sin \alpha \end{Bmatrix} + \sum_{i=1}^{n-1} [\mathbf{P}]_{i(x,y)} \begin{Bmatrix} \gamma_i \\ \gamma_{i+1} \end{Bmatrix} \quad (4.2.36)$$

where $[\mathbf{P}]_{i(x,y)}$ is the panel coefficient matrix in airfoil coordinates for the velocity induced at the arbitrary point (x, y) by panel i , extending from node i to node $i+1$.

The pressure, p , at any point in the flow-field can be determined from this local fluid velocity. The momentum equation for inviscid, incompressible, irrotational flow requires

$$\frac{1}{2}V^2 + \frac{p}{\rho} + gh = \text{constant} \quad (4.2.37)$$

The variation in pressure with altitude is negligible in the region close to the airfoil. Thus, the pressure is very closely approximated as

$$p = p_\infty + \rho \frac{V_\infty^2 - V^2}{2} \quad (4.2.38)$$

and the pressure coefficient is

$$C_p \equiv \frac{p - p_\infty}{\frac{1}{2}\rho V_\infty^2} = 1 - \frac{V^2}{V_\infty^2} \quad (4.2.39)$$

where

$$V^2 = V_x^2 + V_y^2 \quad (4.2.40)$$

The lift and moment coefficients can be determined by integrating the pressure forces around the surface of the airfoil. However, it is more computationally efficient to compute the lift and moment coefficients directly from the panel vorticity.

The section lift and moment coefficients for the airfoil can be computed by adding the contributions from each of the $n-1$ panels. The Kutta-Joukowski law gives the lift per unit span generated by each of these panels,

$$\tilde{L} = \rho V_\infty \Gamma = \rho V_\infty \int_{\xi=0}^l \gamma(\xi) d\xi \quad (4.2.41)$$

The section moment that this lift force produces about the origin of the x - y coordinate system is

$$\tilde{m}_{le} = -\rho V_\infty \int_{\xi=0}^l x(\xi) \cos(\alpha) \gamma(\xi) d\xi - \rho V_\infty \int_{\xi=0}^l y(\xi) \sin(\alpha) \gamma(\xi) d\xi \quad (4.2.42)$$

The contribution to the lift and moment coefficients can be expressed from Eqs. (4.2.41) and (4.2.42) as

$$\tilde{C}_L \equiv \frac{\tilde{L}}{\frac{1}{2}\rho V_\infty^2 c} = \frac{2}{V_\infty c} \int_{\xi=0}^l \gamma(\xi) d\xi \quad (4.2.43)$$

and

$$\tilde{C}_{m_{le}} \equiv \frac{\tilde{m}_{le}}{\frac{1}{2}\rho V_\infty^2 c^2} = -\frac{2}{V_\infty c^2} \left[\int_{\xi=0}^l x(\xi) \cos(\alpha) \gamma(\xi) d\xi + \int_{\xi=0}^l y(\xi) \sin(\alpha) \gamma(\xi) d\xi \right] \quad (4.2.44)$$

For any panel, the airfoil coordinates, x and y , as well as the vortex strength, γ , are all linear functions of the axial coordinate, ξ . Thus we can write

$$x(\xi) = x_i + \frac{x_{i+1} - x_i}{l_i} \xi \quad (4.2.45)$$

$$y(\xi) = y_i + \frac{y_{i+1} - y_i}{l_i} \xi \quad (4.2.46)$$

$$\gamma(\xi) = \gamma_i + \frac{\gamma_{i+1} - \gamma_i}{l_i} \xi \quad (4.2.47)$$

Using Eq. (4.2.47) in Eq. (4.2.43), the section lift coefficient contribution from panel i is

$$\tilde{C}_{Li} = \frac{2}{V_\infty c} \int_{\xi=0}^{l_i} \left(\gamma_i + \frac{\gamma_{i+1} - \gamma_i}{l_i} \xi \right) d\xi = \frac{l_i}{c} \frac{\gamma_i + \gamma_{i+1}}{V_\infty} \quad (4.2.48)$$

The section lift coefficient for the entire airfoil is the sum of that induced by all of the $n-1$ vortex panels,

$$\tilde{C}_L = \sum_{i=1}^{n-1} \frac{l_i}{c} \frac{\gamma_i + \gamma_{i+1}}{V_\infty} \quad (4.2.49)$$

Similarly using Eqs. (4.2.45) through (4.2.47) in Eq. (4.2.44), the section moment coefficient contribution from panel i is

$$\begin{aligned} \tilde{C}_{m_{le}i} &= -\frac{2}{V_\infty c^2} \left[\cos(\alpha) \int_{\xi=0}^{l_i} \left(x_i + \frac{x_{i+1} - x_i}{l_i} \xi \right) \left(\gamma_i + \frac{\gamma_{i+1} - \gamma_i}{l_i} \xi \right) d\xi \right. \\ &\quad \left. + \sin(\alpha) \int_{\xi=0}^{l_i} \left(y_i + \frac{y_{i+1} - y_i}{l_i} \xi \right) \left(\gamma_i + \frac{\gamma_{i+1} - \gamma_i}{l_i} \xi \right) d\xi \right] \\ &= -\frac{1}{3} \frac{l_i}{c} \left[\frac{2x_i\gamma_i + x_i\gamma_{i+1} + x_{i+1}\gamma_i + 2x_{i+1}\gamma_{i+1}}{cV_\infty} \cos(\alpha) + \frac{2y_i\gamma_i + y_i\gamma_{i+1} + y_{i+1}\gamma_i + 2y_{i+1}\gamma_{i+1}}{cV_\infty} \sin(\alpha) \right] \end{aligned} \quad (4.2.50)$$

and the section moment coefficient for the entire airfoil is,

$$\begin{aligned} \tilde{C}_{m_{le}} &= -\frac{1}{3} \sum_{i=1}^{n-1} \frac{l_i}{c} \left[\frac{2x_i\gamma_i + x_i\gamma_{i+1} + x_{i+1}\gamma_i + 2x_{i+1}\gamma_{i+1}}{cV_\infty} \cos(\alpha) \right. \\ &\quad \left. + \frac{2y_i\gamma_i + y_i\gamma_{i+1} + y_{i+1}\gamma_i + 2y_{i+1}\gamma_{i+1}}{cV_\infty} \sin(\alpha) \right] \end{aligned} \quad (4.2.51)$$

Table 4.2.1 shows how the predicted section lift and moment coefficients vary with the number of nodes used to synthesize the airfoil. Typically adequate results are obtained using cosine clustering with 50 to 100 nodes.

Table 4.2.1. The effect of grid refinement on the lift and moment predicted by the vortex panel method.

n	NACA 2412		NACA 2406	
	$\tilde{C}_L$	$\tilde{C}_{m_{c/4}}$	$\tilde{C}_L$	$\tilde{C}_{m_{c/4}}$
10	0.239838	-0.0519393	0.224353	-0.0505011
20	0.255466	-0.0549336	0.239146	-0.0535885
50	0.260138	-0.0557432	0.243221	-0.0544573
100	0.260844	-0.0558644	0.243871	-0.0545891
150	0.260968	-0.0558714	0.243978	-0.0546216
200	0.260987	-0.0558931	0.244050	-0.0546294

Summary

An airfoil of arbitrary shape can be synthesized using $n-1$ vortex panels placed between n nodes on the airfoil surface. The first node is placed at the trailing edge. The nodes are spaced forward along the lower surface, wrapped up around the leading edge, and then spaced back along the upper surface to the trailing edge. The strength of each vortex panel is assumed to be linear along the panel.

The 2×2 **panel coefficient matrix in airfoil coordinates**, $[\mathbf{P}]_{i(x,y)}$, for the velocity induced at the arbitrary point (x,y) by panel i , extending from the node at (x_i, y_i) to the node at (x_{i+1}, y_{i+1}) , is computed from the algorithm:

$$\begin{aligned}
 l_i &= \sqrt{(x_{i+1} - x_i)^2 + (y_{i+1} - y_i)^2} \\
 \begin{Bmatrix} \xi \\ \eta \end{Bmatrix} &= \frac{1}{l_i} \begin{bmatrix} (x_{i+1} - x_i) & (y_{i+1} - y_i) \\ -(y_{i+1} - y_i) & (x_{i+1} - x_i) \end{bmatrix} \begin{Bmatrix} (x - x_i) \\ (y - y_i) \end{Bmatrix} \\
 \Phi &= \text{atan2}(\eta l_i, \eta^2 + \xi^2 - \xi l_i) \\
 \Psi &= \frac{1}{2} \ln \left[\frac{\xi^2 + \eta^2}{(\xi - l_i)^2 + \eta^2} \right] \\
 [\mathbf{P}]_{i(x,y)} &= \frac{1}{2\pi l_i^2} \begin{bmatrix} (x_{i+1} - x_i) & -(y_{i+1} - y_i) \\ (y_{i+1} - y_i) & (x_{i+1} - x_i) \end{bmatrix} \begin{bmatrix} [(l_i - \xi)\Phi + \eta\Psi] & (\xi\Phi - \eta\Psi) \\ [\eta\Phi - (l_i - \xi)\Psi - l_i] & (-\eta\Phi - \xi\Psi + l_i) \end{bmatrix}
 \end{aligned}$$

The $n \times n$ **airfoil coefficient matrix**, $[\mathbf{A}]$, is generated from the 2×2 panel coefficient matrix in airfoil coordinates, $[\mathbf{P}]_{ji}$, for the velocity induced at control point i by panel j , extending from node j to node $j+1$, using the algorithm:

$$\begin{aligned}
 A_{ij} &= 0., \quad i=1, n; \quad j=1, n \\
 \left. \begin{aligned} A_{ij} &= A_{ij} + \frac{x_{i+1} - x_i}{l_i} P_{21ji} - \frac{y_{i+1} - y_i}{l_i} P_{11ji} \\ A_{ij+1} &= A_{ij+1} + \frac{x_{i+1} - x_i}{l_i} P_{22ji} - \frac{y_{i+1} - y_i}{l_i} P_{12ji} \end{aligned} \right\}, \quad i=1, n-1; \quad j=1, n-1 \\
 A_{n1} &= 1. \\
 A_{nn} &= 1.
 \end{aligned}$$

The n **nodal vortex strengths**, γ_1 through γ_n , are then obtained by solving the $n \times n$ linear system,

$$[\mathbf{A}] \begin{Bmatrix} \gamma_1 \\ \gamma_2 \\ \vdots \\ \gamma_{n-1} \\ \gamma_n \end{Bmatrix} = V_\infty \begin{Bmatrix} [(y_2 - y_1) \cos \alpha - (x_2 - x_1) \sin \alpha] / l_1 \\ [(y_3 - y_2) \cos \alpha - (x_3 - x_2) \sin \alpha] / l_2 \\ \vdots \\ [(y_n - y_{n-1}) \cos \alpha - (x_n - x_{n-1}) \sin \alpha] / l_{n-1} \\ 0. \end{Bmatrix}$$

Once the nodal strengths are known, the **velocity and pressure at any point in space** can be computed by adding the velocity induced by all $n-1$ vortex panels to the free-stream velocity,

$$\begin{Bmatrix} V_x \\ V_y \end{Bmatrix} = V_\infty \begin{Bmatrix} \cos \alpha \\ \sin \alpha \end{Bmatrix} + \sum_{i=1}^{n-1} [\mathbf{P}]_{i(x,y)} \begin{Bmatrix} \gamma_i \\ \gamma_{i+1} \end{Bmatrix}$$

$$V^2 = V_x^2 + V_y^2$$

$$C_p \equiv \frac{p - p_\infty}{\frac{1}{2} \rho V_\infty^2} = 1 - \frac{V^2}{V_\infty^2}$$

The **lift and moment coefficients for the entire airfoil** are the sum of those induced by all of the $n-1$ vortex panels,

$$\tilde{C}_L = \sum_{i=1}^{n-1} \frac{l_i}{c} \frac{\gamma_i + \gamma_{i+1}}{V_\infty}$$

$$\begin{aligned} \tilde{C}_{m_{le}} = & -\frac{1}{3} \sum_{i=1}^{n-1} \frac{l_i}{c} \left[\frac{2x_i \gamma_i + x_i \gamma_{i+1} + x_{i+1} \gamma_i + 2x_{i+1} \gamma_{i+1}}{c V_\infty} \cos(\alpha) \right. \\ & \left. + \frac{2y_i \gamma_i + y_i \gamma_{i+1} + y_{i+1} \gamma_i + 2y_{i+1} \gamma_{i+1}}{c V_\infty} \sin(\alpha) \right] \end{aligned}$$

Comparison with Experimental Data

Section lift and moment coefficients predicted by thin airfoil theory and panel codes are in good agreement with experimental data for low Mach numbers and small angles of attack. In Fig. 4.2.5, the inviscid lift coefficient for a NACA 2412 airfoil, as predicted by thin airfoil theory, is compared with the inviscid lift coefficient predicted by the vortex panel method and with experimental data for total lift coefficient as reported by Abbott and Von Doenhoff (1949). Thin airfoil theory predicts a section lift coefficient that is independent of the thickness distribution and dependent only on angle of attack and camber line shape. At small angles of attack, the thin airfoil approximation agrees closely with experimental data based on total lift, for airfoils as thick as about 12 percent. The agreement seen in Fig. 4.2.5 is quite typical. For airfoils much thicker than about 12 percent, viscous effects become increasingly important, even at fairly low angles of attack, and the inviscid lift coefficient predicted by thin airfoil theory begins to deviate more from experimental observations based on total lift. This can be seen in Fig. 4.2.6.

The reason why thin airfoil theory predicts total lift so well over such a wide range of thickness is that thickness tends to increase the lift slope slightly, while viscous effects tend to decrease the lift slope. Since both thickness effects and viscous effects are neglected in thin airfoil theory, the resulting errors tend to cancel, giving the theory a broader range of applicability than would otherwise be expected. Coincidentally, because of these opposing errors, thin airfoil theory actually predicts a lift slope that agrees more closely with experimental data for total lift than does that predicted by the more elaborate vortex panel method. The vortex panel method accurately predicts the pressure distribution around the airfoil, including the effects of thickness. However, since the vortex panel method provides a potential flow solution, it does not account for viscous effects in any way.

After seeing that thin airfoil theory predicts total section lift better than the vortex panel method, one may wonder why we should ever be interested in the vortex panel method. For the answer, we must remember that both the vortex panel solution and the thin airfoil solution come from potential flow theory. Thus, neither of these two solutions can be expected to predict viscous forces. However, potential flow solutions are often used as boundary conditions for viscous flow analysis. Potential flow is used to predict the pressure distribution around the airfoil. The viscous forces are then computed from boundary layer theory. However, since boundary layer flow is greatly affected by the pressure distribution around the airfoil, our potential flow solution must accurately predict the section pressure distribution. Because thin airfoil theory does not account for airfoil thickness, it cannot be used to

predict the surface pressure distribution on an airfoil section with finite thickness. The vortex panel method, on the other hand, accurately predicts the inviscid pressure distribution for airfoils of any thickness.

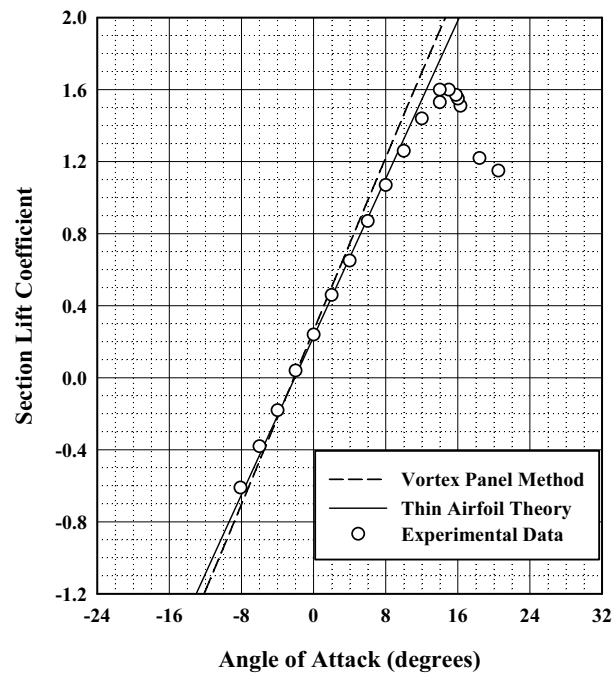


Figure 4.2.5. Section lift coefficient comparison among thin airfoil theory, the vortex panel method, and experimental data based on total lift, for the NACA 2412 airfoil.

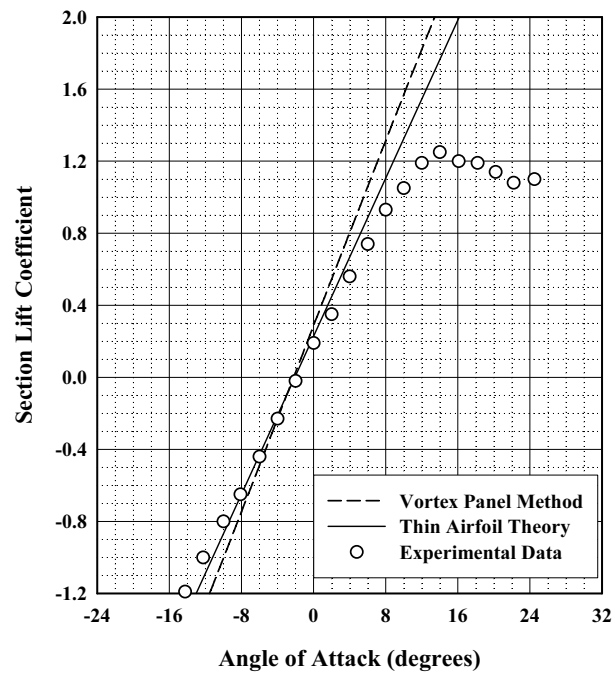


Figure 4.2.6. Section lift coefficient comparison among thin airfoil theory, the vortex panel method, and experimental data based on total lift, for the NACA 2421 airfoil.

While the experimental data shown in Figs. 4.2.5 and 4.2.6 are based on total section lift, section lift data have been obtained that are based on pressure forces only. This is usually accomplished by spacing a large number of static pressure taps around the circumference of an airfoil. The net pressure force acting on the airfoil section is derived by numerically integrating the forces obtained from these pressure measurements. The lift coefficient predicted by thin airfoil theory and that predicted by the vortex panel method are compared with experimental data of this type in Fig. 4.2.7. Notice that the vortex panel method agrees very closely with these experimental data, while thin airfoil theory predicts a lift coefficient that is somewhat low.

In summary, thin airfoil theory can be used to obtain a first approximation for the total section lift coefficient, at small angles of attack, produced by airfoils of thickness less than about 12 percent. Thin airfoil theory gives no prediction for the section drag.

To improve on the results obtained from thin airfoil theory, we can combine a boundary layer solution with the velocity and pressure distribution obtained from the vortex panel method. Since this analytical procedure accounts for both thickness and viscous effects, it produces results that agree closely with experimental data for total lift and drag over a broad range of section thickness. Another alternative available with today's high-speed computers is the use of computational fluid dynamics (CFD). However, a CFD solution will increase the required computation time by several orders of magnitude.

In Figs. 4.2.5 through 4.2.7 it should be noticed that at angles of attack near the zero-lift angle of attack, the result predicted by thin airfoil theory agrees very well with that predicted by the vortex panel method and with that observed from all experimental data. Thus, we see that the thickness distribution has little effect on the lift produced by an airfoil at angles of attack near the zero-lift angle of attack. The thickness distribution does, however, have a significant effect on the maximum lift coefficient and on the stall characteristics of the airfoil section. This is seen by comparing Fig. 4.2.5 with Fig. 4.2.6. The two airfoils described in these two figures have exactly the same camber line. They differ only in thickness. Notice that the NACA 2412 section has a maximum lift coefficient of about 1.6, while the NACA 2421 airfoil produces a maximum lift coefficient of only about 1.2. Also notice that the thinner section has a sharper and more abrupt stall than that displayed by the thicker section.

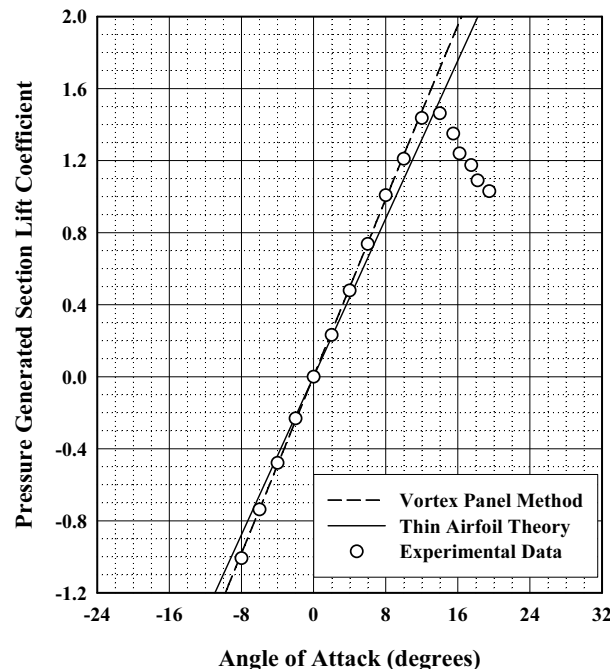


Figure 4.2.7. Section lift coefficient comparison among thin airfoil theory, the vortex panel method, and experimental data based on pressure lift, for the NACA 0015 airfoil.